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Ziemba

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[54] **PHOSPHATE COATING COMPOSITION
AND METHOD**

3,269,877 8/1966 Schlossberg et al. 148/6.16

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FOREIGN PATENTS OR APPLICATIONS

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[51] **Int. Cl.²** **C23F 7/08**

[58] **Field of Search** 148/6.15 R, 6.15 Z, 6.17,
148/6.24, 6.16

[57] **ABSTRACT**

A novel, improved phosphate coating composition and method are provided for overcoming the inhibitory effect on coating caused by the presence of a nitrite agent in the phosphate coating composition.

[56] **References Cited**

UNITED STATES PATENTS

3,181,976 5/1965 Yager..... 148/6.15 Z

2 Claims, No Drawings

PHOSPHATE COATING COMPOSITION AND METHOD

BACKGROUND OF THE INVENTION

Field of the Invention

This invention pertains to the treatment of ferrous metal surfaces with a phosphate composition to enhance the appearance and corrosion resistance thereof by the deposition of a coating thereupon. Generally, an immersion process is utilized, whereby the ferrous metal article to be treated is placed in a phosphate bath for a period of time sufficient to allow deposition of a suitable coating thereupon.

More specifically, the invention relates to an improved zinc phosphate composition and method for improving the coating action of a phosphate coating bath which has been inhibited by the presence of a nitrite agent.

BACKGROUND OF THE INVENTION

It is well known that zinc phosphate compositions (hereinafter also referred to as "phosphate compositions") can be most useful in providing improved appearance and corrosion resistance to ferrous metal surfaces which are treated therewith. U.S. Pat. No. 3,269,877 (Schlossberg et al.) provides such a zinc phosphate coating composition and is hereby incorporated by reference.

Generally, the ferrous metal article is immersed in a bath containing an aqueous phosphate coating composition for a sufficient period of time to allow deposition of a coating of predetermined thickness, usually on the order of 2,000 mgs. per sq. ft., for an acceptable level of salt spray protection.

Difficulties, however, have been experienced when for no apparent reason a normal working bath fails to deposit an adequate coating, or may even stop coating almost entirely. As a result, the corrosion resistance as measured by the salt spray performance of the coating, the thickness of which may be on the order of 1,000 mgs. per sq. ft. or less, is quite poor.

It is thought that this difficulty in achieving an adequate coating thickness is caused by the presence of a nitrite agent, such as nitrogen-oxygen compositions, which for some reason form in the phosphating bath and act as very strong inhibitors to coating action. Not only is it then necessary to counteract the inhibitory effect on the phosphating bath caused by the nitrite agent, but it is also important to do so in a manner which does not adversely effect the stability of the zinc phosphating bath, or the properties and quality of the coating deposited on the ferrous metal surface being treated.

Attempts have been made in the past to alleviate the inhibitory effect of nitrites on working phosphate coating baths. For example, peroxy-sulfur compounds or reducing sulfur-oxygen compounds, such as thiosulfate, have been introduced into a phosphating bath in order to control the level of nitrite or at least negate its inhibitory effect. However, these addition agents are highly unstable and it is not generally feasible, for practical purposes, to prepare a concentrated phosphate coating bath containing these chemicals. Moreover, any excess addition of these sulfur-oxygen additives may adversely effect the bath balance and result in poor coating action and/or the formation of sludge.

SUMMARY OF THE INVENTION

A novel, improved phosphate coating composition and method are provided in accordance with the present invention for overcoming the inhibitory effects on the coating action of a phosphate coating composition caused by the presence of a nitrite agent therein. For purposes of the present invention, the term "nitrite agent" is understood to include any compound having one or more nitrite groups contained therein, as well as oxides of nitrogen, including nitric oxide and nitrogen dioxide, and complexes formed between these oxides of nitrogen and metals found in the phosphate coating composition.

In accordance with the present invention, an agent is provided in, or added to, a zinc phosphating composition to prevent, or mitigate, adverse inhibitory effect of a nitrite agent which may be present or which may develop therein. Urea, addition products of urea, such as urea nitrate, sulfamic acid, alkali metal salts of sulfamic acid, such as sodium or potassium sulfamate, ammonium sulfamate, ascorbic acid and hydroxylamine derivatives, such as hydroxylamine hydrochloride, are provided by the present invention as novel, effective, and useful agents for counteracting the inhibitory effects of a nitrite agent and may be included in an improved phosphate coating composition, or added to a phosphate coating composition which is inhibited in its coating action.

Therefore, it is an object of the present invention to provide an improved phosphate coating composition for producing a phosphate coating on ferrous metal surfaces which will maintain optimum coating action and remain otherwise unaffected in performance by the presence of nitrite therein.

It is a further object of this invention to provide an improved phosphate coating composition containing an agent which is effective to overcome the adverse effects on the coating of ferrous metal surfaces caused by the presence of a nitrite agent therein, while not adversely affecting the performance or coating of the improved phosphate coating composition.

It is yet a further object of this invention to provide an improved phosphate coating composition which is stable in storage in the liquid concentrations used to provide an aqueous solution required for a working zinc phosphate bath.

It is also an object of this invention to provide a method for improving the coating action of an inadequately coating zinc phosphate bath, the coating action of which is inhibited by the presence of a nitrite agent therein.

It is yet a further object of the present invention to provide a method of counteracting the inhibitory effects caused by the presence of a nitrite agent in a phosphate coating bath, without adversely effecting either the performance and coating action of the same, as measured by the rate of deposition of an adequate coating weight, or the appearance and corrosion resistance of the deposited coating.

Other objects and advantages of the present invention will be readily apparent to those skilled in the art in light of the following description of the preferred embodiments of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

The present invention provides a new and improved phosphate coating composition containing an agent

which is stable and effective in alleviating or counteracting the inhibitory effect on coating action caused by the presence of a nitrite agent therein. Furthermore, in accordance with the present invention a novel method is provided to prevent, or relieve, the adverse inhibitory effect of a nitrite agent which may be present or which may form in a phosphate coating composition, wherein an agent for counteracting the inhibitory effects of a nitrite agent on coating action is added to the phosphate coating composition.

I have discovered that urea, addition products of urea, such as urea nitrate, sulfamic acid, alkali metal or ammonium salts of sulfamic acid, such as ammonium sulfamate, ascorbic acid and hydroxylamine derivatives, such as hydroxylamine hydrochloride, are effective as agents for counteracting the inhibitory effects of a nitrite agent, in accordance with the present invention. All of these agents are stable when incorporated into the concentrated phosphate composition used to make-up a "working bath" solution for application of a phosphate coating. Furthermore, even if a large excess of these agents is present in a working bath, no objectionable precipitates or noxious gases are produced, whereby these agents are safe and flexible in their use.

The term "phosphate coating composition", as it is used throughout this specification and in the appended claims, refers to any chemical composition which contains, at a bare minimum, zinc, phosphate, and nitrate ions, and which is effective for depositing a coating thereof upon a ferrous metal surface to which the same is applied. Of course, various additional ingredients may be present in a phosphate coating composition, and various methods of application may be used including, but not limited to, immersion, spraying, brushing and the like. Furthermore, the term "coating action" is generally used herein in reference to the application of a coating of a phosphate coating composition to a fer-

manner. First, the 2 × 4 inch steel panels were immersed in a cleaner solution at 180°F. for at least 10 minutes, so that the ferrous metal surfaces would be uniformly prepared. After rinsing, these panels were immersed in a 4%, by volume, solution of Detrex 502 zinc phosphate coating composition at 200°F. for a period of 15 minutes. The Detrex 502 solution is an aqueous zinc phosphate coating composition comprised essentially of a water solution containing phosphate, nitrate, zinc cobalt, and ferrous ions. These panels were then rinsed and blown dry with air.

The coating weight deposited on these panels was obtained by first weighing the dry treated panels, then removing the coating by treatment with a 25%, by volume, solution of sodium hydroxide and finally reweighing the clean, dry panels. The difference in weight noted between the coated panel and the stripped panel, multiplied by a factor of 9 equals the coating weight in mg. per sq. ft.

Thereafter, the above procedure was followed in various test samples of the Detrex 502 working bath, in which incremental amounts of sodium nitrite (NaNO_2), in a range from 0.2 to 2.0 g/l were added. These modified Detrex 502 solutions were kept hot for 4 hours, then allowed to cool to room temperature overnight and finally re-heated before making final measurements. This temperature cycle roughly approximated the commercial operation of phosphate coating baths in accordance with this invention.

Finally, an amount of urea in the range of about 0.1 to 10.0 g/l was added to each of the various samples of the Detrex 502 working baths. Results of this experiment are presented in Table 1, below, and show the marked improvement in coating action of the working bath after addition of urea. Furthermore, the improvement in coating was also visually evident in the appearance of the deposited coating.

Table 1

Coating Weight over time (hrs) in mg/ft ²	NaNO ₂ (g/l)					
	0	0.2	0.5	1.0	1.5	2.0
0 hr	1870	503	495	472	415	324
1 hr	1295	445	472	530	630	652
18 hr	785	1230	1000	822	828	895
19* hr	2180	2760	1450	1565	1410	1640

*after adding 1 g/l of urea

rous metal surface, or the surface of an alloy of the same.

Preferred embodiments of the composition and method of the present invention are illustrated in the following examples, which are not intended to be limiting in any manner.

EXAMPLE 2

The procedure of Example 1, above, was repeated, except that the various samples were allowed to stand over a weekend before making final measurements. The results, appearing in Table 2, below, again indicate the improvement of coating action after addition of urea.

Table 2

Coating Weight over time (hrs) in mg/ft ²	NaNO ₂ (g/l)					
	0	0.2	0.5	1.0	1.5	2.0
66 hr	740	910	836	622	670	610
67* hr	2780	1460	1690	1530	2050	2640

*after adding 1 g/l of urea

EXAMPLE 1

In order to demonstrate the inhibitory effect of a nitrite agent when present in an immersion zinc phosphating bath and the alleviation of this inhibition by the addition of urea, mild steel panels were subjected to the coating action of a zinc phosphate bath in the following

EXAMPLE 3

To study the stability of urea in Detrex 502 concentrates, the following compositions were prepared:

Sample No.	1	2	3	4	Blank
% urea	0.5	1.0	2.0	2.5	0
% Detrex 502	99.5	99.0	98.0	97.5	100

Samples of these various compositions were then stored for a period of 480 hours, after which each sample was analyzed to determine the % urea remaining. The following results were obtained:

Ambient Temperature				
Sample No.	1	2	3	4
% urea added	0.5	1.0	2.0	2.5
% urea found	0.38	0.79	1.46	1.90

130°F.					
Sample No.	1	2	3	4	Blank
% urea added	0.5	1.0	2.0	2.5	0
% urea found	0.37	0.81	1.41	1.91	0.13

These results indicate that within the experimental error of the method of analysis which was used, there was no significant decomposition of urea, nor any indication of reaction between nitrate and urea.

EXAMPLE 4

A standard Detrex 502 concentrate was compared with one containing 1% urea to further demonstrate the stability of urea therein. Each of these materials was placed in a 125 ml. flask and stoppered with a one-holed rubber stopper through which a 1 ml. Mohr pipette was inserted. The stoppers were pressed down until each pipette was at least half full of concentrate solution.

Any decomposition of urea would be accompanied with gas evolution, which when generated would cause the liquid level in one of these pipettes to rise more than the other.

Over a storage period of three weeks, at ambient temperature, no such evolution of gas or increase in liquid level in either of the pipettes was noted.

EXAMPLE 5

In order to determine the minimum amount of urea required to counteract the effect of a nitrite agent in a Detrex 502 bath, as well as to determine the effect of large amounts of urea upon the coating action of the same, the method of preparation, treatment and coating weight determination in Example 1 were repeated and the following variations prepared:

Normal 4%, by volume, Detrex 502 Solution

g/l urea added	0	1	10	50
coating weight in mg/ft ²	2280	2500	2710	2260

This procedure was repeated using the 4% Detrex 502 bath, modified with 0.2g/l sodium nitrite, and left standing for 4 hours at 200°F. The following results are representative:

4% Detrex 502, with 0.2g/l NaNO ₂				
g/l urea added	0	0.2	1.0	2.0
coating weight in mg/ft ²	1175	2300	2190	2480

A standard solution, tested in accordance with the procedure followed above, but without additional sodium nitrite or urea produced a coating weight of 1960 mg/ft² after 5 hours.

EXAMPLE 6

Following the procedure of Example 1, steel panels and a Detrex 502 bath, modified with a 0.1 g/l of sodium nitrite, were prepared. After addition of neutralized sulfamic acid, followed by an additional hour of standing, the following coating weights were determined:

g/l sulfamic acid added	0	0.2	0.5	1.0	1.5	2.0
coating weight in mg/ft ²	1090	2290	2470	2510	2380	2470

Furthermore, to determine the amount of sulfamic acid which can be tolerated by a Detrex 502 bath, without adversely effecting the coating action thereof, various amounts of sulfamic acid neutralized with ammonium hydroxide were added to the Detrex 502 bath, with the following results:

g/l sulfamic acid added	0	1	10
coating weight in mg/ft ²	2220	2390	2170

EXAMPLE 7

To study the stability of sulfamates in the Detrex 502 concentrate, a solution was prepared from 25 g sulfamic acid in water with about 23 ml NH₄OH(30%NH₃). This solution was then diluted to exactly 50 ml.

The following dilutions of the above solution with Detrex 502 were made on a weight basis:

Percent Sulfamate Solution	1.0	2.0	4.0	10.0
Percent Detrex 502	99.0	98.0	96.0	90.0

These solutions were divided into 2 equal samples, placing one in an oven at 130°F while the other was maintained at room temperature for 500 hrs. No evolution of gas or formation of precipitate was noted in any sample.

EXAMPLE 8

To determine the effectiveness of other agents in counteracting the inhibition of a nitrite agent, the procedure of Example 1, wherein a Detrex 502 bath was modified with 0.7 g./l. of sodium nitrite, was repeated and 1 g/l of each of the following counteracting agents was added. The results are summarized in Table 2, below:

Table 2

Counteracting Agents	Coating Weight in mg/ft ²
Na ₂ S ₂ O ₃	774
Mannitol	504
Formaldehyde	482
Diethanolamine	567
Ethanolamine	477
Dextrose	437
NH ₂ OH HCl	2116
Ascorbic Acid	2349
Cyclohexylamine	423
Paraformaldehyde	414
Urea Nitrate	2066
Aniline	629

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It is therefore evident that hydroxylamine hydrochloride, ascorbic acid, and urea nitrate are also effective in counteracting the inhibitory effect of a nitrite agent in a Detrex 502 bath phosphate coating solution.

It will be readily apparent to those skilled in the art that various modifications may be made in the details of the method of the present invention, such as the sequence or increments in which the counteracting agent is added to the zinc phosphating bath, and also that other ingredients may be present in a phosphate coating composition of the invention, all within the spirit and scope of the invention as recited in the appended claims.

I claim:

1. A method of preventing inhibition of effective coating action of an aqueous zinc phosphate coating bath on ferrous metal surfaces due to the presence of

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nitrite ions in such coating bath, said method comprising:

- a. providing an aqueous zinc phosphate coating bath,
- b. adding sodium nitrite to said zinc phosphate coating bath, thereby forming nitrite ions, and
- c. adding to said bath a compound selected from the group consisting of urea and urea nitrate in an amount in the range from about 0.1 grams per liter to about 10 grams per liter of the total solution.

2. The method of claim 1 wherein said aqueous zinc phosphate coating bath consists essentially of an aqueous solution containing a phosphate ion-producing agent, a nitrate ion-producing agent, a zinc ion-producing agent, a cobalt ion-producing agent, and a ferrous ion-producing agent.

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